

Problem 1: Put 6 counters in piles of 2, 2, 1 and 1. A move goes like this: take one counter from every pile, and put those counters together as one new pile. Make moves, and after each move write the pile sizes in the next row, biggest first. Stop when a row is exactly the same as the row above it. How many moves did it take until the piles stopped changing?

Move	Piles, biggest first
Start	2 2 1 1
1	
2	
3	
4	
5	
6	
7	

Moves until it stopped: _____

Problem 2: Do the same from three more starts: two piles of 3; one pile of 6; six piles of 1. Write the number of moves for each start.

Move	Piles, biggest first
Start	3 3
1	
2	
3	
4	
5	

Moves from 3 3: _____

Move	Piles, biggest first
Start	6
1	
2	
3	
4	
5	

Moves from 6: _____

Move	Piles, biggest first
Start	1 1 1 1 1 1
1	
2	
3	
4	
5	

Moves from 1 1 1 1 1 1: _____

Problem 3: There are exactly 11 ways to arrange 6 counters in piles. They are all listed here. Next to each one, write how many moves it takes until the piles stop changing. You already know four of them; run the rest with your counters.

6	_____	3 1 1 1	_____
5 1	_____	2 2 2	_____
4 2	_____	2 2 1 1	_____
4 1 1	_____	2 1 1 1 1	_____
3 3	_____	1 1 1 1 1 1	_____
3 2 1	_____		

Problem 4: Now 8 counters. Put all 8 in one pile. Make moves and write each row. Stop when a row is the same as an earlier row, not just the row above it. Draw a bracket beside the rows that form the loop. How many arrangements are in the loop?

Move	Piles, biggest first
Start	8
1	
2	
3	
4	
5	
6	
7	
8	

Arrangements in the loop: _____

Problem 5: Start 8 counters as four piles of 2, and then as piles of 5, 1, 1, 1. For each start, make moves and write rows until a row repeats an earlier row. Which of your three 8-counter games go round on the same arrangements?

Move	Piles, biggest first
Start	2 2 2 2
1	
2	
3	
4	
5	

Move	Piles, biggest first
Start	5 1 1 1
1	
2	
3	
4	
5	
6	

Same loop as the game from one pile of 8: _____

Problem 6: Here are all 22 arrangements of 8 counters. Share them out at your table. Run each one until it repeats an arrangement, and write L or S next to it: L if it ends in the long loop of four arrangements, S if it ends in the short loop of two. Count how many arrangements end in each loop.

8	_____	3 3 2	_____
7 1	_____	3 3 1 1	_____
6 2	_____	3 2 2 1	_____
6 1 1	_____	3 2 1 1 1	_____
5 3	_____	3 1 1 1 1 1	_____
5 2 1	_____	2 2 2 2	_____
5 1 1 1	_____	2 2 2 1 1	_____
4 4	_____	2 2 1 1 1 1	_____
4 3 1	_____	2 1 1 1 1 1 1	_____
4 2 2	_____	1 1 1 1 1 1 1 1	_____
4 2 1 1	_____		
4 1 1 1 1	_____		

Long loop: _____ arrangements

Short loop: _____ arrangements

Problem 7: Run 4 counters from one pile of 4, and 5 counters from one pile of 5, until a row repeats an earlier row. Then list all seven arrangements of 5 counters in the space on the right. Use your list to explain why every game with 5 counters must repeat an arrangement by move 7 at the latest.

Move	Piles, biggest first
Start	4
1	
2	
3	
4	
5	

Move	Piles, biggest first
Start	5
1	
2	
3	
4	
5	
6	

Problem 8: Run 10 counters from three starts: one pile of 10; two piles of 5; five piles of 2. Stop each run when a row is the same as the row above it, and write the number of moves it took.

Move	Piles, biggest first
Start	10
1	
2	
3	
4	
5	
6	
7	

Moves from 10: _____

Move	Piles, biggest first
Start	5 5
1	
2	
3	
4	
5	
6	
7	

Moves from 5 5: _____

Move	Piles, biggest first
Start	2 2 2 2 2
1	
2	
3	
4	
5	

Moves from 2 2 2 2 2: _____

Problem 9: Run 3 counters from three piles of 1, 7 counters from one pile of 7, and 9 counters from piles of 5 and 4, each until a row repeats an earlier row. Then fill in the table for every number of counters from 3 to 10, using all your runs so far: does the game stop, or go round? If it goes round, how many arrangements are in each loop you found?

Move	Piles, biggest first
Start	1 1 1
1	
2	
3	

Move	Piles, biggest first
Start	7
1	
2	
3	
4	
5	
6	
7	

Move	Piles, biggest first
Start	5 4
1	
2	
3	
4	
5	

Counters	Stops, or goes round?	Arrangements in each loop found
3		
4		
5		
6		
7		
8		
9		
10		

Problem 10: Find an arrangement of 10 counters that does not change at all when you make a move. Is there a second one? Then try to find an arrangement of 8 counters that does not change under a move. Explain why there is none: if a move changes nothing, which pile must be the new pile, and what does that force the other piles to be?

Problem 11: The game stops for 3, 6 and 10 counters, at the stairs 2–1, 3–2–1 and 4–3–2–1. Write the next three numbers of counters for which you expect the game to stop, and the stairs it should stop at. Test the first of them: put that many counters in five piles of 3 and run until a row is the same as the row above it.

Next stopping numbers: _____, _____, _____ Stairs they stop at: _____

Move	Piles, biggest first
Start	3 3 3 3 3
1	
2	
3	
4	
5	
6	

Moves until it stopped: _____

Problem 12: Undo a move. Find every arrangement that turns into 3, 2, 1 after one move. Find every arrangement that turns into 5, 3, 2 after one move. Then try to find one that turns into 2, 2, 2, 2.

Turn into 3 2 1: _____

Turn into 5 3 2: _____

Turns into 2 2 2 2: _____

Problem 13: Sort these arrangements of 8 counters: which can be reached by a move from some arrangement, and which cannot? For each one that can, write an arrangement that reaches it. Then write what all the unreachable ones have in common, in a way that lets you decide for 6, 1, 1 and for 2, 2, 2, 1, 1, 1, 1 without running anything.

Arrangement	Reached from	Cannot be reached
4 4		
3 3 1 1		
2 2 2 2		
2 2 2 1 1		
5 1 1 1		
3 2 1 1 1		
3 1 1 1 1 1		
4 2 1 1		

Problem 14: Find a start with 10 counters that takes more than 10 moves to reach the stairs. Write each start you try in a table and record its run.

Move	Piles, biggest first
Start	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	

Move	Piles, biggest first
Start	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	

Problem 15: Build the stairs 3–2–1 with 6 counters. Add one extra counter to the pile of 3 and write the piles, biggest first, in the first blank. Move the extra counter to the pile of 2 instead and write that; then to the pile of 1; then put it down as a new pile of 1. Compare your four arrangements with the loop for 7 counters in Problem 9. Then write all five arrangements that are the stairs 4–3–2–1 with one extra counter (11 counters). Start 11 counters at 5, 3, 2, 1 and make 6 moves. Does the game stay inside your list? Finally, write all six arrangements that are the stairs 3–2–1 with one extra counter on each of two different steps (a new pile of 1 counts as a step), and compare them with the two loops from Problems 4 and 5.

Stairs 3–2–1 plus one: _____

Move	Piles, biggest first
Start	5 3 2 1
1	
2	
3	
4	
5	
6	

Stairs 4–3–2–1 plus one: _____

Stairs 3–2–1 plus two: _____

Compared with the loops for 8 counters:

Problem 16: 12 counters. Write all ten arrangements that are the stairs 4–3–2–1 with one extra counter on each of two different steps. Predict how many loops there are for 12 counters and which arrangements are in each. Then test: run 5, 4, 2, 1 for 6 moves and run 5, 3, 3, 1 for 6 moves. Are they in the same loop? Find a start outside your list that ends in the same loop as 5, 3, 3, 1.

Move	Piles, biggest first
Start	5 4 2 1
1	
2	
3	
4	
5	
6	

Move	Piles, biggest first
Start	5 3 3 1
1	
2	
3	
4	
5	
6	

Same loop? _____ A start outside the list, in the loop of 5 3 3 1: _____

Problem 17: How many moves does it take from 5, 5, 5 to reach the stairs 5–4–3–2–1? And from 4, 4, 3, 2, 1, 1? Record each run.

Move	Piles, biggest first
Start	5 5 5
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	
14	
15	
16	
17	
18	

Move	Piles, biggest first
Start	4 4 3 2 1 1
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	
14	
15	
16	
17	
18	
19	
20	
21	

Moves from 5 5 5: _____

Moves from 4 4 3 2 1 1: _____